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Social sustainability and digital transformation - two birds with one stone? A concept paper for a mixed methods research design

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Abstract

Digital transformation processes have been challenging organizations for a while. Moreover, sustainability has added another objection, aiming at reaching the United Nations Sustainable Development Goals as well as fulfilling environmental social governance reporting standards. Sustainability includes economic, ecological and social dimensions. However, previous research mainly focused on the first two aspects indicating contradictory results, especially in connection with digital transformation processes. Currently, the EU-27 countries struggle to include social sustainability in the digital transformation process design. The guiding question for this research is whether social sustainability is taken into account in digital transformation processes and how these affect employees regarding their motivation and health. The purpose of this concept paper is to outline a mixed method research design of a project using method triangulation. First, a systematic review combines prior literature on social sustainability in digital transformation processes in the areas of business psychology and information systems. Second, a two-wave online survey using the job demands-resources model examines the effects of the individual perception of social sustainability in digital transformation processes and digital skills on work engagement and digital transformation stress, adding job crafting and technical affinity as a mediator. Third, a vignette scenario study consisting of a 2x2 matrix includes digital transformation and social sustainability to high and low extent and analyses the effect within the previously stated adaption of the job demands-resources model, however, including technical affinity as a mediator. Consequently, in the sense of a triple transformation, the studies add the social aspect as a third dimension to the current twin transformation discussion referring to digital and ecological (green) transformation. This research project building on social sustainability in digital transformation processes demands for interdisciplinary work in designing studies and choosing appropriate triangulated methods to identify scientifically sound measures to practical complex challenges. Moreover, it is among the first projects comparing current literature in information systems and business psychology. The results of the studies should not only contribute to the theoretical and practical discussion, but also derive an interdisciplinary research agenda for social sustainability in digital transformation processes.

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1. Introduction

Digital transformation (DT) processes have been a challenge for organizations of all kinds for some time [1]. The emergence of digital technologies such as smart technology, artificial intelligence (AI) or automation, robotics, cloud computing, and the Internet of Things accelerated these changes [24]. Not only technological innovation, but also constantly changing regulatory or customer requirements as well as the retention or winning of qualified employees, remain a competitive advantages in rapidly changing times [2; 24]. This fundamentally changes the way people work and how companies function [24]. These changes involve benefits and disadvantages for employees [11]. Employees are forced to use new workplace technologies and work in new structures due to the constant adaption of the business model [5]. Not only is the content of the work increasingly digitalized, but social interaction and communication have also become increasingly digitalized as the international or remote collaborations or departments replace the traditional human services by advanced AI tools [5]. Changes in organizations can be conceived from top down, but also bottom up initiatives foster those processes [7]. Stress caused by organizational changes interferes with the integration of the DT process, but little academic research has been conducted to understand DT process as a demand for employees' health and motivation [11]. These changes are combined with concerns about the future of jobs and companies [24]. The research question for the current research project is therefore: *How do people perceive DT processes, as well as themselves, regarding their health and motivation at work?*

1.1. Digital transformation

Over the past decade, digital technology has transformed society and the global economy. Many companies continue to drive their digital transformation in order to meet the needs of their stakeholders and customers [17]. DT processes are embedded in increasing business efficiency and effectiveness for example using AI or machine learning to update the business model or add services to the portfolio [2]. These processes require changes in the nature of work [5]. Although DT is a widely accepted term, there is a lack of definitions and established measurements of the organizational digital readiness is missing or in proposition [24; 29]. DT has to be distinguished from the organizational transformation processes of information technology (IT). The first redefines the organizational value proposition for the customers, leading to a new organizational identity, and the second is to improve the existing identity through increased use of digital technology [28].

In DT processes, success depends not only on the strategy and its operationalization, but also on employee adaption and support [24]. The transformation process itself includes opportunities and risks for all parties involved, such as the organization itself, its leaders and employees [5]. Specifically, for employees, new working models are combined with new working environments, new technologies, and changes in tasks and routines [11]. These changes do not only involve resources but also challenges for employees [11]. Consequently, both the resilience of employees and active dealing with the effect of challenging the individual well-being are required [24]

1.2. Sustainability and social sustainability in digital transformation processes

The Brundtland Commission on Environment and Development defines sustainability as the use of resources that ensure future generations' possibility to do the same [6]. Thus, sustainability includes both economic, ecological and social dimensions [1]. Adding sustainability to the discussion of digital technologies, the term of digital sustainability originated from the concept of digital content enabling, preserving, encouraging, and technologizing digital content in enterprises. [29]. Social sustainability (SOS) refers to the implications of technology and digitalization on society as a whole in the process of DT including potential adverse effects on work and private life [17]. SOS is defined as the preservation and advancement of the well-being and quality of life of individuals and communities [1]. It emphasizes ensuring the equitable distribution and preservation of social benefits and opportunities over time.

However, the relationship between DT and sustainability was little studied in the past (interdisciplinary) studies, with little attention to the social aspects of sustainability that still require definitions and conceptual understanding [2; 15]. Previous research focused on economic and ecological aspects, while only in recent years, attention has been paid to research social sustainability at the global and sectoral levels. This research, draws attention to the equal importance of the three dimensions of sustainability [1; 18].

As a result of the growing importance of organizations in achieving sustainability objectives, not only digital transformation, but also the existence of a sustainable business model, is no choice, but a must have criterion for the success of the future business and improvement of efficiency [2]. On country level, Finland, the Netherlands and Denmark show highest values in Europe with regards to adding social aspects to DT processes increasing not only those processes itself but also showing higher performances in reaching the SDGs [18]. Despite the collective attention to economic and ecological needs for sustainability, fewer research included the importance of the social aspect [2; 18].

Prior research indicates positive effects adding the SDG principles as a design factor to the DT process for increasing success and outcomes [2]. Examining the effect of social constructs, there is an indication of influence on all the aspects of sustainability for organizations [20]. However, literature shows contradictory results for the direction of the interdependency between (social) sustainability and DT [2; 18]. The direction of causal effects needs further investigation. Explanations can be assumed in the complex and challenging nature of social constructs, especially SOS, but also increasing well-being, individual development or innovativeness [20]. In changing times and a rapidly increasing demand for employees' digital skills, there is a chance for employees to learn using new platforms making work life easier [2].

1.3. Job Demands-Resources model

Since DT processes itself and additionally including SOS include challenges and opportunities for all parties involved, the underlying theoretical model needs to respect those two sides of the coin. Defining organizational change processes as demands affecting individuals, the current project uses the job demands-resources model (JDR) including individual demands and resources as challenges and assets of work situations in the context of DT processes [3; 11]. The model shows not only the two sides but also the effects on employees' health and motivation making it suitable for the current context [4]. Previous data indicated the model being heuristically usable for redesigning work places reducing work related stress on the one hand, and fostering work engagement on the other hand [8]. In contrast to the possibility of using the JDR for work groups, departments or on an organizational level, this research projects uses it to assess the individual perception of the DT and the individual resources for the health and motivational effects [3; 8]. Empirical findings suggest that work related demands influence employees' health as well as their detachment from work. In contrast, work related resources are beneficially related to engaging in work [8]. On the individual level, research suggests five essential factors to coping with DT processes in a healthy manner:

- technology literacy
- personal perceptions and attitudes towards technological change
- individual skills and professional training
- resilience
- work-related adaptability,
- and work-related well-being [24]

In the context of DT processes, prior results suggest the importance of digital skills as suitable resources [18]. This is why this research project focusses on those individual skills. Building up on former research on the health related effect of burnout, instead, the influence on DT related stress (DTS) is added to the model in the context of DT processes [14; 4]. Adding job crafting as an opportunity to design the current job role, the model outlines how individuals design their work situation within the given constraints. As, prior research additionally indicates, job crafting reduces suffering from work-related stress and increases ongoing motivation [4]. Moreover, job crafting enhances the individual meaning of one's job being stimulative by suitable work environment conditions, e. g. fostered by change [7].

In order to outline the above-mentioned approach, the paper includes an overview of the methods and materials used for all three studies. This section consists of an overview of the systematic review first, followed by the two-wave online survey and the two times two scenario study. The paper ends with a conclusion.

2. Methods and material

Triangulated mixed methods research projects provide a comprehensive approach to answering the underlying research question, mitigating the flaws of the individual method [26]. Prior research in SOSDT included different methodological approaches. However, studies mainly focused on single sectors or disciplines raising demand in interdisciplinary and mixed method approaches [1]. Data integration in conceptualization as well as analysis is essential in designing and conducting research projects suitable to real life complexity including qualitative and quantitative approaches [9].

The contrasting state of literature as well as the missing insights in the current state of employees' perception of SOSDT processes and little prior scenario studies in this area indicate the need for a method triangulation [9; 18; 10]. This research projects includes an ethics vote as well as a pre-registration contributing to open science standards. The following part outlines the theoretical and practical reasoning for designing a mixed methods research project.

2.1. Systematic Review

Despite the raising interest and importance of DT processes and sustainability, prior researchers raise attention to the ongoing need for concept clarification of SOS being multidimensional [15; 20]. Academic contribution also lacks further effort to derive a common understanding of variables and relations in this area [10]. Therefore, the first research project conducts a systematic review using the preferred reporting items for systematic reviews and meta-analyses (PRISMA) as a guideline [19]. This adds knowledge to the current state of literature by providing an overview of existing research [20]. Moreover, it adds to the need of further comparative approaches in understanding the interdisciplinary topic of SOSDT processes [28]. With this, the research questions of the first projects state: *What variables are used in the field of information systems as compared to the field of business psychology on the topic of digital, socially sustainable transformation? How can the variables found be categorized?* Therefore, the inclusion criteria are defined as follows:

- relevant articles from peer-reviewed journals
- assigned to the fields of business psychology or information systems
- primary data or in the form of a systematic review
- written in English or German
- published in the period 2015 – 2024
- fully accessible
- attributable to the research question
- articles that do not fulfil the inclusion criteria or that deal exclusively with economic or ecological aspects are excluded
- duplicates if not already excluded by the criteria above

Besides the above-mentioned inclusion criteria, information sources and the search strategy include Science Direct, AIS eLibrary, Scopus, WoS as well as EBSCOhost (Academic Search Complete, Business Source Premier, APA PsycInfo). Some search term include the German terminology. The search term for title, abstract and key words include ("digital* transformation" OR "digitalisation" OR "digitalization" OR "Twin Transformation") AND ("social sustainability" OR "sustainability" OR Nachhaltigkeit OR "soziale Nachhaltigkeit" OR "responsib"). After data choice and extraction, the study quality evaluation succeeds the data analyses and syntheses done by three researchers.

The insights of the systematic review do not only add scientific contribution by collecting but combining the current state of research in business psychology and information systems. With this, a future research roadmap can be designed added by insights in employees' perception of the current digital transformation processes and its' effects on their health and motivation as included in the next study.

2.2. Two wave online survey

Contrasting existing academic contribution to DT insights on the organizational level, fewer studies provide insights in individual perceptions [16]. Prior insights given point to a lack of digital competencies leading to inequality and exclusion and may jobs today requiring at least a basic level of digital literacy. In addition, the promotion of digital literacy contributes to the development of the digital economy. Moreover, promoting digital literacy also contributes to workforce development, improves access to information and promotes entrepreneurship by reducing digital inequality [18]. Additionally, higher digital competencies are related to lower digital stress levels being trainable by qualified support in the work context [14]. Due to existing mediation and moderation effects on job crafting, this study includes both mediation and moderation in the following way: Fig. 1 visualizes the JDR model underlying the current research project added by the above-argued variables.

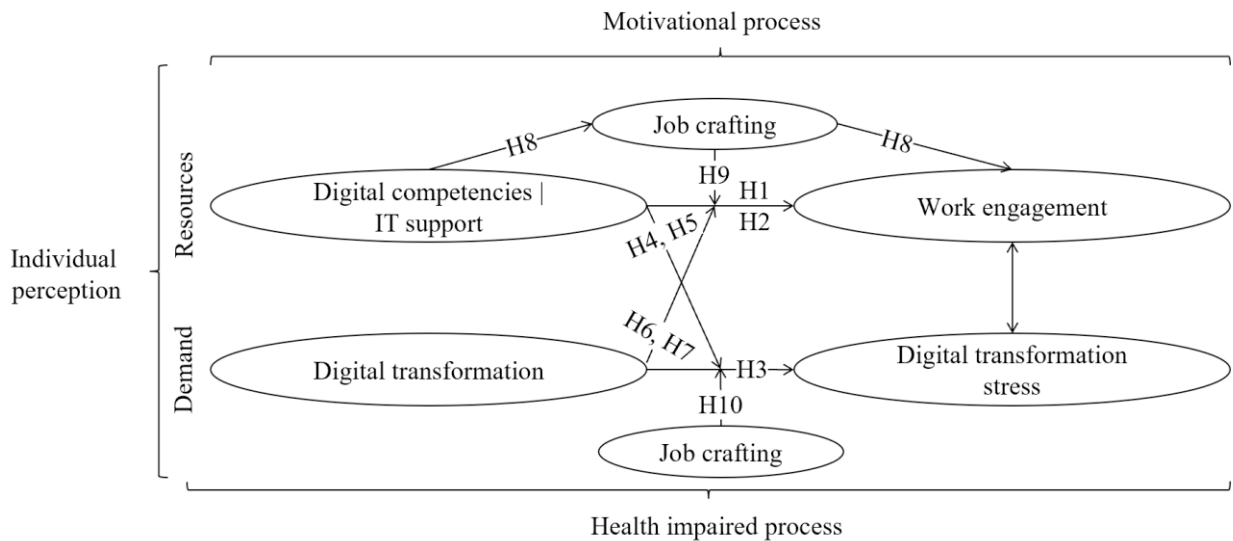


Fig. 1. Research model for the two wave online survey based on the JDR in the context of SOSDT processes including job crafting as a mediator and moderator, own illustration based on adaption to existing models [3; 4]

The research questions in this projects state: *How are digital transformations perceived by employees in terms of work engagement and digital transformation stress, considering employees' digital competencies and IT support from the organization? Does job crafting mediate the path between digital competencies and work engagement? Does job crafting moderate the relationship between digital competencies and work engagement, and digital transformation and digital transformation stress?* The visualized research design to the research questions adds the following hypotheses to this project part:

- H1: There is a positive relationship between digital competencies and work engagement.
- H2: There is a positive relationship between IT support and work engagement.
- H3: There is a positive relationship between digital transformation and digital transformation stress.
- H4: The positive relationship between digital transformation and digital transformation stress is moderated by digital competencies, such that the relationship is lower when digital competences are high (vs. low).
- H5: The positive relationship between digital transformation and digital transformation stress is moderated by IT support such that the relationship is lower when IT support is high (vs. low).
- H6: The positive relationship between digital competencies and work engagement is moderated by digital transformation, such that it is higher when digital transformation is high (vs. low).

- H7: The positive relationship between IT support and work engagement is moderated by digital transformation, such that it is higher when digital transformation is high (vs. low).
- H8: Job crafting mediates the positive relationship between digital competencies and work engagement.
- H9: The positive relationship between digital competencies and work engagement is increased by job crafting. Thus, the relationship between digital competencies and work engagement is higher for employees who have a high (vs. low) level of job crafting.
- H10: The positive relationship between digital transformation and digital transformation stress is reduced by job crafting. Thus, the relationship between digital transformation and digital transformation stress is lower for employees who have a high (vs. low) level of job crafting.

The data collection will take place in two waves separating dependent and independent variables with a time lag of two weeks aiming to achieve 200 completed data sets. All the planned scales for the variables use a five-point Likert-scale. To assess the information and communications technology (ICT) skills, the *digital competences* scale including five items is used [23]. *Technical support* includes three items [21]. *DTS* includes six items [14]. The variable *work engagement* is included with nine items using a short scale (Utrecht Work Engagement Scale-9,) [22]. Additional items assess job crafting where item reduction is done based on discriminatory power [13; 12]. Control variables are age, gender, working hours per week, the number of employees and the organizational department. For data evaluation, the first step is to clean and prepare the data from the survey for analysis. Participants who do not meet the requirements or have incomplete or missing values are excluded. The time taken to answer the questions will also be checked. People whose response times are much shorter than average may be excluded if necessary. Afterwards, testing for the assumptions of normal distribution, homoscedasticity, multicollinearity and outliers for a linear regression analysis will be carried out. Next, descriptive statistics will be analyzed. Moderation and mediation analyses are planned as part of the inferential statistical analyses.

For exploratory purposes, the following analyses are made:

- The relationship between digital competencies and digital transformation stress.
- The relationship between IT support and digital transformation stress.
- The relationship between digital transformation and work engagement.
- The relationship between digital transformation and digital transformation stress mediated by job crafting.
- Effect of the separate dimensions of job crafting (increasing structural resources, increasing social resources, increasing challenging demands, reducing hindering demands) and their moderating and mediating effects on the relationship between digital competencies and work engagement as well as between digital transformation and digital transformation stress.

The insights of this research design add to the current state of employees' perception on digital transformation processes and its health impairing and motivational effects. This adds to the state of research and provides a starting point for practical measures in the future. Depending on the status of the digital transformation and the degree to which social sustainability is taken into account, suitable measures can be derived

2.3. Two times two scenario study

Previous research indicated positive results for individuals and organizations for experimental study design using the JDR[8]. Due to the need of adding employee interest to the SOS and DT discussion, this study includes technical affinity as a moderating variable to the JDR [20; 18]. Fig. 2 visualizes the JDR model underlying the current research project added by the above-argued variables.

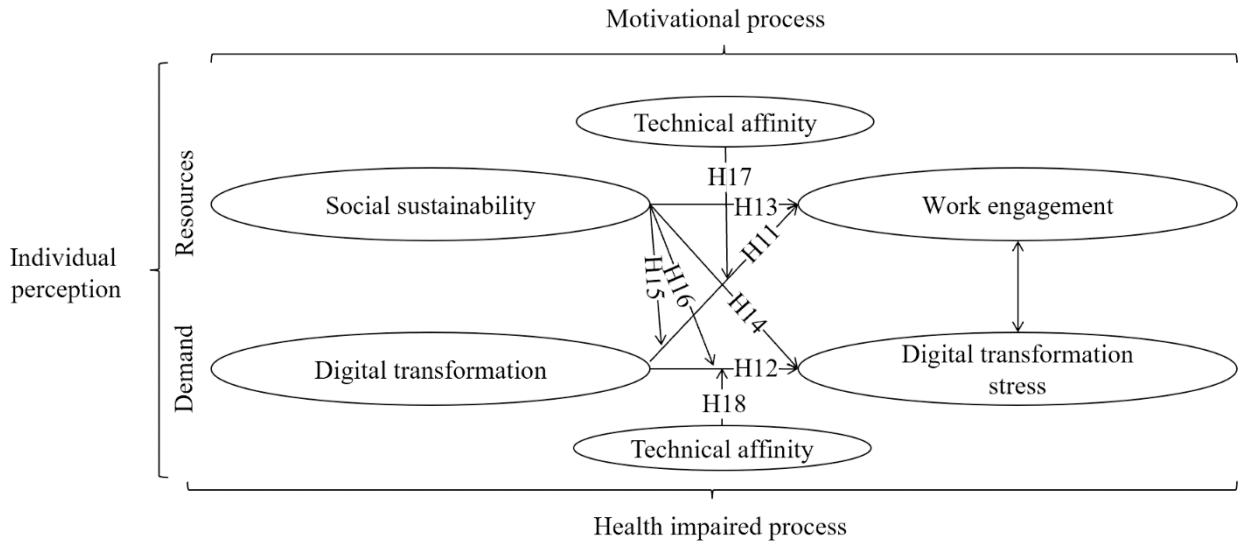


Fig. 2. Research model for the two times two scenario study based on the JDR in the context of SOSDT processes including technical affinity as a moderator, own illustration based on adaption to existing models [3; 4]

For adding a quasi-experimental approach assessing scenarios with high and low extent of DT and SOS as a two times two scenario study, the third project part includes the following research questions: *How does socially sustainable digital transformation affect engagement and mental health at work? And how does respondents' technical affinity affect the results?* These questions result in the following hypotheses based on prior research for the first part focusing on the extent of digital transformation processes and social sustainability (high vs. low):

- H11: A high level of digital transformation leads to lower work engagement.
- H12: A high level of digital transformation leads to a higher level of stress among employees.
- H13: A high level of social sustainability leads to higher work engagement.
- H14: A high level of social sustainability leads to a lower level of stress among employees.
- H15: Social sustainability moderates the relationship between digital transformation and work engagement in such a way that the relationship is stronger when social sustainability is high.
- H16: Social sustainability moderates the relationship between digital transformation and stress among employees in such a way that the relationship is weaker when social sustainability is high.
- H17: Technical affinity moderates the relationship between digital transformation and work engagement in such a way that the relationship is stronger when technical affinity is high.
- H18: Technical affinity moderates the relationship between digital transformation and the experience of stress at work in such a way that the relationship is reduced when technical affinity is high.

The data collection will take place online randomly and equally distributing participants to one of the four scenarios aiming at 200 completed data sets. The planned scales for the variables and control variables are similar to those in the two wave study adding technical affinity with nine items [27]. The hypotheses of the study assume main effects as well as moderating effects. The first step of data evaluation is to clean the data. As the scenarios are categorical variables, dummy variables have to be created. Subsequently, the assumptions of normal distribution, homoscedasticity, multicollinearity and outliers for a linear regression analysis will be checked. Both a regression analysis for the main effects of the vignettes on employees' stress experience and work engagement and a regression analysis including the moderator technology affinity are to be carried out. Finally, a multivariate cluster-corrected regression is planned for statistical analysis.

The third study offers scientific insights by adding an experimental scenario study, including employees' perceptions of high and low levels of digital transformation and social sustainability at work and thus increasing internal validity. Not only does the method itself contribute to a third approach to address the overall research question, but the findings also provide practical opportunities for action in all possible phases of organizational development of digital transformation processes.

3. Conclusion

This project aims to derive practical insights in the current state as well as levers to improve employees' perception of DT processes based on the JDR-model [25]. This research project's rationale outlines the theoretical and practical need in deriving a research design to address the current state of literature as well as measuring the employees' health and motivational effects of SOSDT processes and individual resources. The results of the systematic review are expected to show specific needs of research, especially in the area of business psychology. The insights given in the two-wave online survey assumably indicate motivational and health related effects of the current perception of DT process designs. And finally, the scenario study results might show an internally valid effect of SOSDT on employees' health and motivation. The mixed methods approach aims to mitigate the limitations of each method by providing a triangulated approach in answering the question of how individuals perceive DT processes and SOS as well as their own digital skills. Those actionable insights enhance not only employees' well-being but may also support organizational performance [25]. With this, the current twin transformation research approach is enhanced to adding more importance to the third dimension of sustainability, making it a triple transformation process.

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